Factorization and Rank

MATRIX FACTORIZATIONS

A *factorization* of a matrix A is an equation that expresses A as a product of two or more matrices. Whereas matrix multiplication involves a *synthesis* of data (combining the effects of two or more linear transformations into a single matrix), matrix factorization is an *analysis* of data.

LU Factorization of Matrix

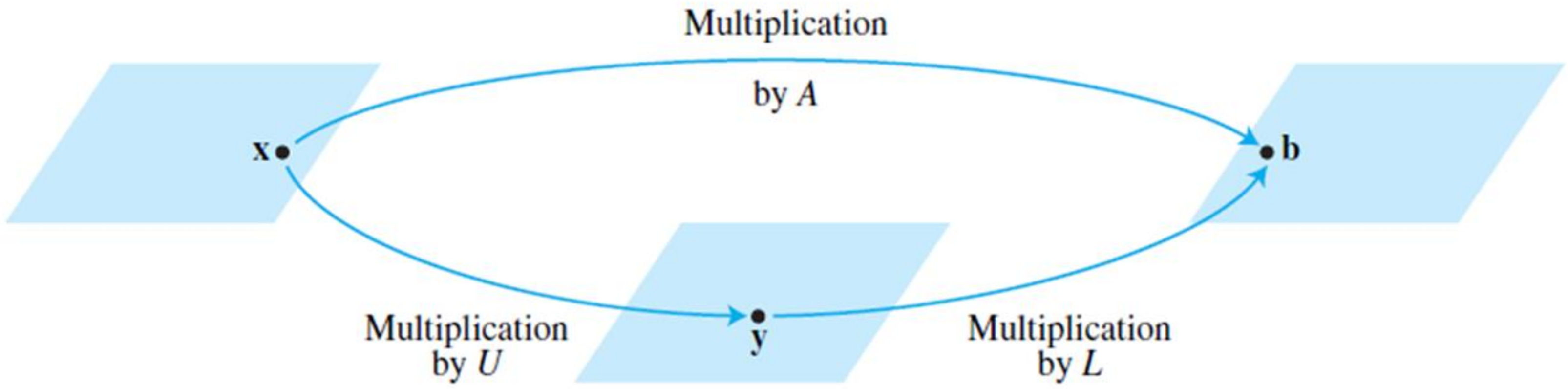
LU factorization is a way of decomposing a matrix A into an upper triangular matrix U , a lower triangular matrix L and a permutation matrix P such that PA = LU .

The solution to a system Ax = b of linear equations can be solved quickly if A can be factored as A = LU. In this section we show that gaussian elimination can be used to find such factorizations.

Before studying how to construct L and U, we should look at why they are so useful. When $A = LU$, the equation $A\mathbf{x} = \mathbf{b}$ can be written as $L(U\mathbf{x}) = \mathbf{b}$. Writing $\mathbf{y}$ for $U\mathbf{x}$, we can find $\mathbf{x}$ by solving the *pair* of equations

$$\boxed{\begin{aligned} L\mathbf{y} &= \mathbf{b} \\ U\mathbf{x} &= \mathbf{y} \end{aligned}}$$

First solve $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$, and then solve $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$. See Figure Each equation is easy to solve because L and U are triangular.



Find an LU factorization of $A = \begin{bmatrix} 1 & 2 & 0 & 2 \\ 1 & 3 & 2 & 1 \\ 2 & 3 & 4 & 0 \end{bmatrix}$.

Solution

One way to find the LU factorization is to simply look for it directly. You need

$$\begin{bmatrix} 1 & 2 & 0 & 2 \\ 1 & 3 & 2 & 1 \\ 2 & 3 & 4 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ y & z & 1 \end{bmatrix} \begin{bmatrix} a & d & h & j \\ 0 & b & e & i \\ 0 & 0 & c & f \end{bmatrix}.$$

Then multiplying these you get

$$\begin{bmatrix} a & d & h & j \\ xa & xd+b & xh+e & xj+i \\ ya & yd+zb & yh+ze+c & yj+iz+f \end{bmatrix}$$

and so you can now tell what the various quantities equal. From the first column, you need $a = 1, x = 1, y = 2$. Now go to the second column. You need $d = 2, xd + b = 3$ so $b = 1, yd + zb = 3$ so $z = -1$. From the third column, $h = 0, e = 2, c = 6$. Now from the fourth column, $j = 2, i = -1, f = -5$. Therefore, an LU factorization is

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 6 & -5 \end{bmatrix}.$$

You can check whether you got it right by simply multiplying these two.

LU Factorization

Pivoting in matrix operations refers to a technique used during matrix factorization (such as LU decomposition) or Gaussian elimination to improve numerical stability and avoid division by zero. It involves rearranging (swapping) rows or columns of the matrix to position the largest absolute value in a specific column as the "pivot element."

LU factorization is a method used to decompose a matrix A into the product of two matric lower triangular matrix L and an upper triangular matrix U, such that:

$$A = LU$$

Here's how it works with an example:

LU Factorization without Pivoting

Given the matrix:

$$A = \begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix}$$

We want to decompose A into L and U.

Step 1: Initialize L and U

We start by assuming that:

$$L = \begin{pmatrix} 1 & 0 \\ l_{21} & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{pmatrix}$$

Step 2: Decompose matrix A

Using the equation $A = LU$, we multiply L and U:

$$\begin{pmatrix} 1 & 0 \\ l_{21} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{pmatrix} = \begin{pmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{pmatrix}$$

Problem 1

Comparing this with $A = \begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix}$, we get the following system of equations:

1. $u_{11} = 4$
2. $u_{12} = 3$
3. $l_{21}u_{11} = 6 \rightarrow l_{21} \times 4 = 6$, so $l_{21} = \frac{6}{4} = 1.5$
4. $l_{21}u_{12} + u_{22} = 3 \rightarrow 1.5 \times 3 + u_{22} = 3$, so $u_{22} = 3 - 4.5 = -1.5$

Step 3: Construct L and U

Now we have:

$$L = \begin{pmatrix} 1 & 0 \\ 1.5 & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 4 & 3 \\ 0 & -1.5 \end{pmatrix}$$

Thus, the LU factorization of matrix A is:

$$A = \begin{pmatrix} 1 & 0 \\ 1.5 & 1 \end{pmatrix} \begin{pmatrix} 4 & 3 \\ 0 & -1.5 \end{pmatrix}$$

Problem 2

LU Decomposition of a 3x3 Matrix

Decompose the matrix A below:

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix}$$

Steps:

1. Decompose A into L and U:

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$$

2. Equating the elements of A with the product of L and U:

$$u_{11} = 2, \quad u_{12} = 1, \quad u_{13} = 1$$

$$l_{21} \cdot 2 = 4 \quad \Rightarrow \quad l_{21} = 2$$

$$l_{31} \cdot 2 = -2 \quad \Rightarrow \quad l_{31} = -1$$

$$2 \cdot 1 + u_{22} = -6 \quad \Rightarrow \quad u_{22} = -8$$

Solution (Continued…)

$$2 \cdot 1 + u_{23} = 0 \quad \Rightarrow \quad u_{23} = -2$$

$$-1 \cdot 1 + 7 = -8 \cdot l_{32} \quad \Rightarrow \quad l_{32} = -\frac{3}{4}$$

$$-1 \cdot 1 + u_{33} = 2 \quad \Rightarrow \quad u_{33} = -1$$

So, the LU decomposition is:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -\frac{3}{4} & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & -1 \end{pmatrix}$$

Problem 3

Decomposition with Partial Pivoting

Let's decompose the matrix A using partial pivoting:

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & -2 & -3 \\ 4 & 2 & 2 \end{pmatrix}$$

Since the first element is zero, we need to perform row swaps to make the matrix suitable for LU factorization. Swap row 1 and row 3:

$$A = \begin{pmatrix} 4 & 2 & 2 \\ 1 & -2 & -3 \\ 0 & 2 & 1 \end{pmatrix}$$

Now, proceed with the LU decomposition:

$$A = \begin{pmatrix} 4 & 2 & 2 \\ 1 & -2 & -3 \\ 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$$

Solve the system for L and U.

Solution (Continued…)

Let's solve the LU decomposition of the matrix A with partial pivoting:

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & -2 & -3 \\ 4 & 2 & 2 \end{pmatrix}$$

Since the first element $A[1,1]$ is zero, we need to swap rows for partial pivoting. We swap row 1 and row 3 to avoid division by zero:

$$A = \begin{pmatrix} 4 & 2 & 2 \\ 1 & -2 & -3 \\ 0 & 2 & 1 \end{pmatrix}$$

Now, we perform the LU decomposition on this modified matrix A.

Steps:

1. **Decompose A into L and U:**

$$A = \begin{pmatrix} 4 & 2 & 2 \\ 1 & -2 & -3 \\ 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$$

Solution (Continued…)

2. **Step-by-step factorization:**

   From the first row of A:

   $$u_{11} = 4, \quad u_{12} = 2, \quad u_{13} = 2$$

   From the second row of A:

   $$l_{21} \cdot u_{11} = 1 \quad \Rightarrow \quad l_{21} = \frac{1}{4}$$

   Substitute l_{21} into the second row to solve for u_{22} and u_{23}:

   $$\frac{1}{4} \cdot 2 + u_{22} = -2 \quad \Rightarrow \quad u_{22} = -2.5$$
   $$\frac{1}{4} \cdot 2 + u_{23} = -3 \quad \Rightarrow \quad u_{23} = -3.5$$

   From the third row of A:

   $$l_{31} \cdot u_{11} = 0 \quad \Rightarrow \quad l_{31} = 0$$

Solution (Continued…)

Solve for l_{32} by substituting into the third row:

$$l_{32} \cdot u_{22} = 2 \quad \Rightarrow \quad l_{32} = \frac{2}{-2.5} = -\frac{4}{5}$$

Finally, solve for u_{33}:

$$-\frac{4}{5} \cdot (-3.5) + u_{33} = 1 \quad \Rightarrow \quad u_{33} = 1 - \frac{14}{5} = -\frac{9}{5}$$

Final Matrices L and U:

The lower triangular matrix L is:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{4} & 1 & 0 \\ 0 & -\frac{4}{5} & 1 \end{pmatrix}$$

The upper triangular matrix U is:

$$U = \begin{pmatrix} 4 & 2 & 2 \\ 0 & -2.5 & -3.5 \\ 0 & 0 & -\frac{9}{5} \end{pmatrix}$$

Problem 4

Use 2 LU factorization of A to solve $A\mathbf{x} = \mathbf{b}$,

Where

$$b = \begin{pmatrix} 5 \\ -2 \\ 9 \end{pmatrix}$$

We need to solve $AX = b$, where $A = LU$. The system becomes:

$$LUX = b$$

We solve this in two steps:

1. Solve $LY = b$ for Y (forward substitution).
2. Solve $UX = Y$ for X (back substitution).

Step 2: Solve $LY = b$

We need to solve:

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -\frac{3}{4} & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \\ 9 \end{pmatrix}$$

This gives the following system of equations:

1. $y_1 = 5$
2. $2y_1 + y_2 = -2 \quad \Rightarrow \quad 2(5) + y_2 = -2 \quad \Rightarrow \quad y_2 = -12$

Solution (Continued…)

3. $-y_1 - \frac{3}{4}y_2 + y_3 = 9 \quad \Rightarrow \quad -(5) - \frac{3}{4}(-12) + y_3 = 9 \quad \Rightarrow \quad -5 + 9 + y_3 = 9 \quad \Rightarrow \quad y_3 = 5$

So, we have:

$$Y = \begin{pmatrix} 5 \\ -12 \\ 5 \end{pmatrix}$$

Step 3: Solve $UX = Y$

Now, solve:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -12 \\ 5 \end{pmatrix}$$

This gives the following system of equations:

1. $-x_3 = 5 \quad \Rightarrow \quad x_3 = -5$
2. $-8x_2 - 2x_3 = -12 \quad \Rightarrow \quad -8x_2 - 2(-5) = -12 \quad \Rightarrow \quad -8x_2 + 10 = -12 \quad \Rightarrow \quad -8x_2 = -22 \quad \Rightarrow \quad x_2 = \frac{22}{8} = \frac{11}{4}$

Solution (Continued…)

3. $2x_1 + x_2 + x_3 = 5 \quad \Rightarrow \quad 2x_1 + \frac{11}{4} - 5 = 5 \quad \Rightarrow \quad 2x_1 + \frac{11}{4} - \frac{20}{4} = 5 \quad \Rightarrow$
$2x_1 - \frac{9}{4} = 5 \quad \Rightarrow \quad 2x_1 = 5 + \frac{9}{4} = \frac{20}{4} + \frac{9}{4} = \frac{29}{4} \quad \Rightarrow \quad x_1 = \frac{29}{8}$

So, the solution is:

For AX=b is:

$$X = \begin{pmatrix} \frac{29}{8} \\ \frac{11}{4} \\ -5 \end{pmatrix}$$

Problem 5

Find an LU factorization of

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

First perform row reduction of A to an echelon form U.

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix} = U$$

$$\begin{bmatrix} 2 \\ -4 \\ 2 \\ -6 \end{bmatrix} \begin{bmatrix} 3 \\ -9 \\ 12 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \begin{bmatrix} 5 \end{bmatrix}$$

Use the highlighted entries to obtain L.

Solution (Continued…)

$$\begin{bmatrix} 2 \\ -4 \\ 2 \\ -6 \end{bmatrix} \begin{bmatrix} 3 \\ -9 \\ 12 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \begin{bmatrix} 5 \end{bmatrix}$$

$$\div 2 \qquad \div 3 \qquad \div 2 \qquad \div 5$$

$$\downarrow \qquad \downarrow \qquad \downarrow \qquad \downarrow$$

$$\begin{bmatrix} 1 & & & \\ -2 & 1 & & \\ 1 & -3 & 1 & \\ -3 & 4 & 2 & 1 \end{bmatrix}, \quad \text{and} \quad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$

Solution (Continued…)

Construct L and U for the matrix A.

$$A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -5 & 1 & 0 \\ -3 & 8 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} = LU$$

Problem 7

Use this LU factorization of A to solve $A\mathbf{x} = \mathbf{b}$, where $\mathbf{b} = \begin{bmatrix} -9 \\ 5 \\ 7 \\ 11 \end{bmatrix}$.

$$\boxed{\begin{matrix} L\mathbf{y} = \mathbf{b} \\ U\mathbf{x} = \mathbf{y} \end{matrix}} \longrightarrow \begin{bmatrix} L & \mathbf{b} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ -1 & 1 & 0 & 0 & 5 \\ 2 & -5 & 1 & 0 & 7 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} I & \mathbf{y} \end{bmatrix}$$

The solution of $L\mathbf{y} = \mathbf{b}$ needs only 6 multiplications and 6 additions

Solution (Continued…)

Use this LU factorization of A to solve $A\mathbf{x} = \mathbf{b}$, where $\mathbf{b} = \begin{bmatrix} -9 \\ 5 \\ 7 \\ 11 \end{bmatrix}$.

$$\boxed{\begin{matrix} L\mathbf{y} = \mathbf{b} \\ U\mathbf{x} = \mathbf{y} \end{matrix}} \longrightarrow \begin{bmatrix} U & \mathbf{y} \end{bmatrix} = \begin{bmatrix} 3 & -7 & -2 & 2 & -9 \\ 0 & -2 & -1 & 2 & -4 \\ 0 & 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 3 \\ 4 \\ -6 \\ -1 \end{bmatrix}$$

Then, for $U\mathbf{x} = \mathbf{y}$, the "backward" phase of row reduction requires 4 divisions, 6 multiplications, and 6 additions.

Rank

To put it simply, the *rank* of the matrix represents the amount of independent columns in the matrix. This number, *r,* is very important when examining a matrix. Let's take the rank of this matrix.

$$\begin{bmatrix} 3 & 5 & 6 \\ 4 & 1 & 8 \\ 5 & -2 & 10 \end{bmatrix}$$

The rank of this matrix is 2. That is because we have *two* independent columns, column 1 and 2. The third column is a multiple of the first column, and thus is dependent. Simply, **r = 2.**

Rank

To find the **rank of a matrix**, follow one of these methods. The **rank** is the number of **linearly independent rows** or **columns** in the matrix. Several methods can be used to determine the rank of a matrix, with some being more common in practice.

Row Echelon Form (REF) or Reduced Row Echelon Form (RREF)

This is one of the most common methods to find the rank of a matrix. Here's how it works:

Steps:

1. **Convert the matrix to Row Echelon Form (REF)** using **Gaussian elimination** or **Gauss-Jordan elimination**.
    1. Apply elementary row operations to convert the matrix to upper triangular form or reduced row echelon form.
    2. These operations include swapping rows, multiplying a row by a scalar, and adding/subtracting one row from another.

2. **Count the number of non-zero rows**. The number of non-zero rows in the REF or RREF is the **rank of the matrix**.

Example:

Consider the matrix A:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Perform row operations to get the matrix into REF:

1. Subtract 4 times the first row from the second row.
2. Subtract 7 times the first row from the third row.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix}$$

Now, subtract 2 times the second row from the third row:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix}$$

This is the **row echelon form**. Since there are **two non-zero rows**, the **rank** of the matrix is $\text{Rank}(A) = 2$.

Practice Problem

Q. Find an LU factorization of A and solve the equation $A\boldsymbol{x} = \boldsymbol{b}$ by using the LU factorization.

$$A = \begin{bmatrix} 2 & -4 & 2 \\ -4 & 5 & 2 \\ 6 & -9 & 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix}$$